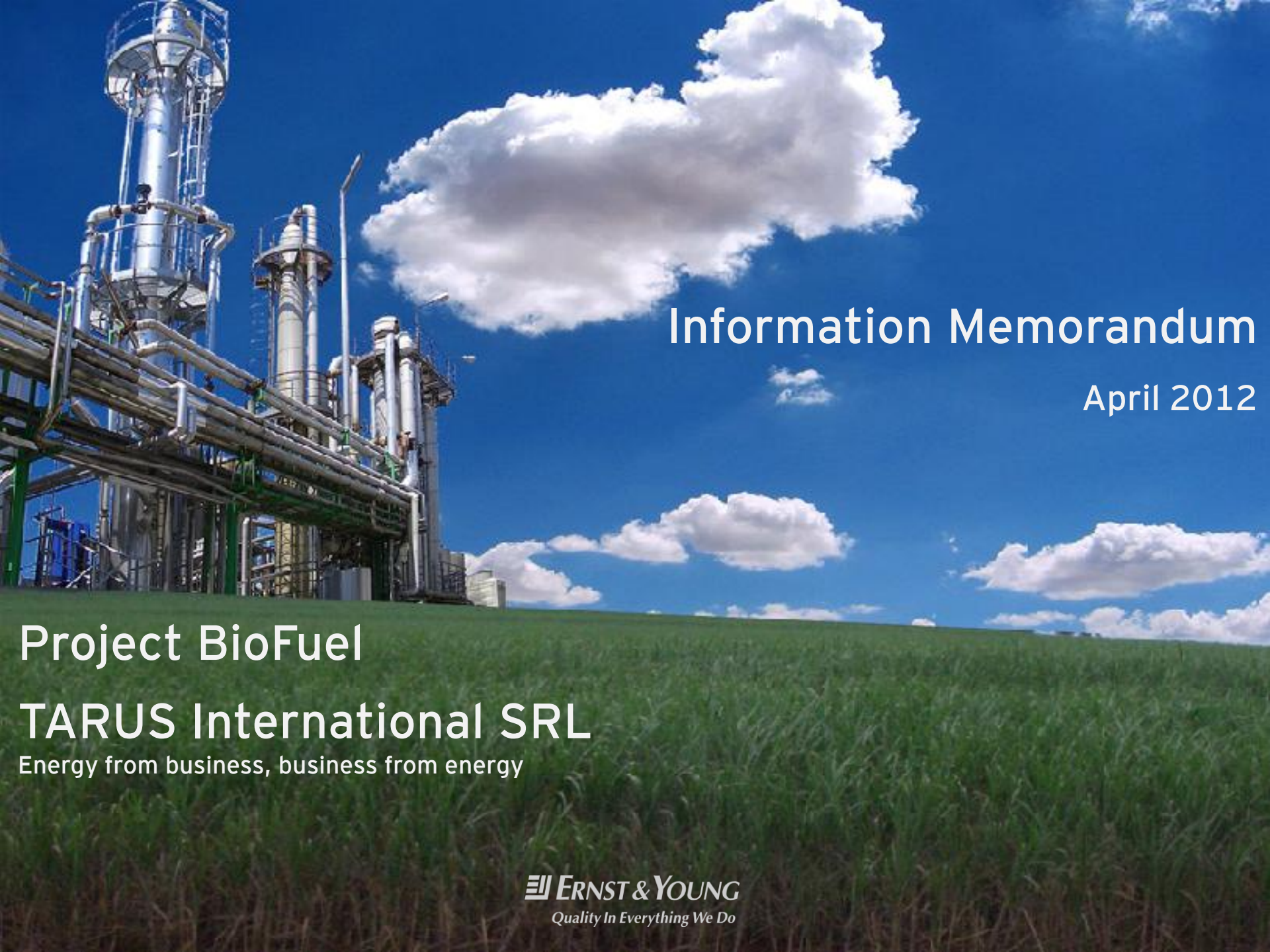




Information Memorandum

April 2012

Project BioFuel

TARUS International SRL

Energy from business, business from energy



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Abbreviations



bn	Billion
BOP	Balance of payment
BP	Business plan
CAPEX	Capital expenditure
Ccy	Currency
CIT	Corporate Income Tax
CRES	Center for Renewable Energy Sources
DDGS	Dried Distillers Grains with Solubles
E10	Fuel mixture that contains 10% ethanol
EBIT	Earnings before interest and tax
EBITDA	Earnings before interest, tax, depreciation and amortization
EtOH	Chemical formula for bioethanol
ETS	Emissions Trading Scheme
EU	European Union
EUR	Euro, the official currency of the European Union
EUR 000	Thousand Euros
FYXXA	Actual financial result in the financial year 20XX
GDP	Gross Domestic Product
GHG	Green House Gasses
GM	General Manager
ha	Hectare
l	liter, liters
hl	Hectoliters
IEA	International Energy Agency
IRR	Internal rate of return
km ²	Square kilometers

Abbreviations



Mb	Million barrels
MDL	Moldovan Leu, the official currency of Moldova
Mha	Million hectares
mn	Million
Mtoe	Million tonnes of oil equivalent
MW	Megawatts
n.a.	Not available
NBM	National Bank of Moldova
NPV	Net present value
NWC	Net working capital
OECD	Organization for Economic Co-operation and Development
p.a.	Per annum
PLDV	Passenger light-duty vehicle
ROCE	Return on capital employed
ROE	Return on equity
t	Tonne
th	Thousand
u.m.	Unit of measurement
VAT	Value added tax
yoy	Year-on-year

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Executive Summary

Transaction rationale

Key investment considerations

Executive Summary

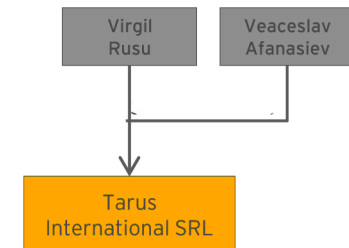
Background and Business overview



Background

- ▶ Tarus International ("Tarus") is a privately owned company, having as objective the construction and operation of a bio-ethanol production facility located near Chisinau, in the Republic of Moldova .
- ▶ The owners are looking for equity partners to support the construction of the bio-ethanol production plant ("the project").
- ▶ Tarus operations will enjoy the following benefits: lower level of operating costs, excellent agricultural environment of Moldova, integrated supply of feedstock as well as the favorable trading climate between Moldova and the EU.
- ▶ The financial projections have been elaborated for a period of 7 years: 2012 - 2019.

Shareholding structure



Business Overview

- ▶ The main areas of activity are:
 - ▶ Sweet sorghum biomass and corn seeds production and processing;
 - ▶ Bioethanol production and sales;
 - ▶ Cogeneration and sale of electricity, DDGS and CO2;
- ▶ The Company's long-term competitiveness is ensured by having in-house experts in the field.
- ▶ Tarus will hold the intellectual property rights for the sorghum hybrids, which, according to the Moldovan law, allows for the exclusive (monopolistic) rights of exploitation.
- ▶ The Moldovan sweet sorghum hybrids ensure a higher productivity per hectare, compared to other crops (no. 16 -20 in Annex 3).
- ▶ Further fiscal advantages resulting from the Company's status as resident of the "ACADEMICA" Science & Technology Park.
- ▶ Based on the trade-off between (1) initial investment needed, (2) further development options and (3) investment recovery time, the production facility has been optimally designed for a capacity of 100 mn liters/year.
- ▶ In addition to the total of 200 Tarus employees, the project will employ, indirectly, circa 220 agricultural workers, for feedstock production.

Main project attributes

Project location	Chisinau, Moldova
Production Capacity	100 mn. liters/year
Raw materials to be used	Sweet sorghum biomass and corn seeds
Number of employees	200
IRR (Equity level)	75.9%
EBITDA margin	min 66%

Executive Summary

Transaction rationale



Transaction rationale

- ▶ Tarus owners are looking for financial support (equity + debt) for the acquisition of the required land for the cultivation of feedstock and the successful completion of the construction phase of the project. The equity partner would enjoy a minority position
- ▶ The current estimated costs of bringing the project to an operational phase are in the range of EUR 131 million (EUR 118 million for equipments and EUR 13 million for working capital).

Executive Summary

Key investment considerations



Area	Attraction
Demand forecast	▶ Biofuels demand is expected to increase rapidly over the projections period, thanks to rising oil prices and government support, prompted by energy-security and environmental concerns; ▶ In the New Policies Scenario (NPS, please refer to Annex 1 for further details), global biofuels consumption increases from 1.1 mb/day today to 4.4 mb/day in 2035. Biofuels are forecasted to meet 8% of world road-transport fuel consumption by 2035, up from 3% in 2009; ▶ According to the NPS, over 2009-2035 biofuels could meet about 20% of global incremental demand for road transport fuels.
Location advantages	▶ Government support (i.e. fiscal advantages – detailed below); ▶ Cheaper labor force (average salary in Moldova is around EUR 250 /month, as compared to an average of EUR 1,800 within the EU); ▶ High soil quality (more than 90% of Moldova’s soil is categorized as chernozem) and availability (around 250,000 ha are not currently exploited) .
Tax incentives	▶ Tax incentives provided to Tarus as resident of the “ACADEMICA” Science & Technology Park: ▶ Exemption from payment of VAT (20%) on imported goods and services for 5 years from start of project; ▶ Exemption from payment of VAT (20%) on goods and services bought on the territory of Moldova for 5 years from start of project; ▶ Exemption from payment of customs taxes (5%) on imported goods and services for 5 years from start of project; ▶ Exemption from payment of income tax (12%) for 3 years from start of operations.
Integrated and efficient production process	▶ Due to the “in-house” supply of sorghum and corn, the Company does not depend on external supply of raw materials;

Key investment considerations cont'd



Area	Attraction
	▶ Further cost reductions through utilization of sorghum as raw material (for full details, please refer to Section 3) : ▶ Up to 3 times more productive in terms of tonnes of ethanol extracted per ha, as compared to other feedstock; ▶ Not demanding on soil fertility, in fact it improves soil fertility by providing positive humus balance; ▶ Sweet sorghum has the highest degree of drought resistance amongst all agricultural crops currently sowed in Europe; ▶ Bears a high degree of soil salinization, and a large interval of soil acidity (pH 4.5-8.5). ▶ During the sorghum-production period (65 days), the company is able to cover its own needs of electric energy (8 MW) as well as sell the surplus (16 MW) on the market. The Renewable Energy law in Moldova stipulates the compulsory acceptance into the national grid, of any volume of electric energy produced from renewable resources]. ▶ By-products obtained during the production process (DDGS and CO₂) that could be sold with attractive margins.
R&D	▶ The internalized R&D team aims to capitalize on their solid experience for: ▶ Genetically improve the sorghum hybrids; ▶ Sowing and harvesting technologies for sweet sorghum and corn; ▶ On-field testing with the most competitive feedstock hybrids across Europe (based on criteria of productivity, quality, as well as resistance to various biotical (parasites) and abiotical (draught, frost) factors.
Environmental impact	▶ Significant reduction of CO₂ footprint for the sorghum cultures, compared to its alternatives; ▶ Sweet sorghum does not require intensive use of fertilizers and water, thus minimizing potential water pollution.

Section 1

Country and Industry analysis

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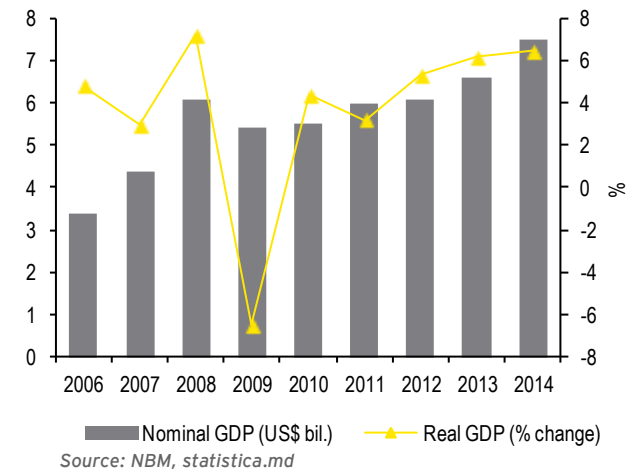
Country overview – The Republic of Moldova



General considerations

- ▶ Located in the north-eastern Balkans, the Republic of Moldova ("Moldova") emerged as an independent republic following the collapse of the USSR in 1991.
- ▶ Moldova occupies an area of 33,843.5 km², and is surrounded by Ukraine (on the North, East and South) and Romania on the West.
- ▶ The total population of Moldova is 3.6 million (UN, 2010).
- ▶ Since the declaration of its independence, Moldova passed through a complex stage of transition to the market economy and experienced a significant economic recession (1990-1999).
- ▶ Over the last decade the economy started to recover, and **between 2000 - 2011 the GDP increased in real terms by 52%**. During 9m11, the GDP increased by 6.7%.
- ▶ Individuals are due to pay an **income tax rate of either 7% or 18% (depending on the income level)**, while companies pay a **corporate income tax of 12%**, one of the **lowest tax rates in Europe**.
- ▶ VAT level is 20%.
- ▶ Some of the main macro-economic indicators of Moldova are detailed in the table below:

Moldova - GDP actual and forecast evolution



Currency: US\$	2006	2007	2008	2009	2010	2011e	2012f	2013f	2014f
Real GDP (% change)	4.8	3	7.2	-6.5	7.1	6.7	5.3	6.2	6.5
Nominal GDP (US\$ bn)	3.4	4.4	6.1	5.4	5.8	6	6.1	6.6	7.5
Nominal GDP Per Capita (US\$)	935	1,229	1,695	1,514	1,530	1,677	1,732	1,869	2,136
Consumer Price Index (% change)	12.7	12.3	12.7	0	7.4	6.6	5.0	4.5	4.1
Fiscal Balance (% of GDP)	-0.3	-0.2	-1	-6.8	-6.1	-3.2	-2.8	-2.6	-1.2
Population (mn)	3.6	3.6	3.6	3.6	3.6	3.6	3.5	3.5	3.5
Unemployment Rate (%)	7.4		4	6.4	10.3	9.9	9.5	9.3	9.1
Current Account Balance (% of GDP)	-11.6	-15.3	-16.3	-8.1	-10.5	-9.1	-8.9	-8.4	-7.5
BOP Exports of Goods US\$ bil	1.1	1.4	1.6	1.3	1.4	1.5	1.7	1.9	2.1
Exchange Rate (MDL/US\$, end of period)	12.9	11.3	10.4	12.3	12.1	11.1	14	14.2	13.2

Source: NBM

Agricultural focus in Moldova

- ▶ Moldova's population has accumulated throughout the years a vast experience and knowledge in growing fruits and vegetables, tobacco, as well as in viticulture and wine production. Consequently, agriculture represents the most important sector of the national economy, using around 27% of country's labor force.
- ▶ Agricultural output accounts for 8.5% of GDP and together with the processing industry represents around 20% of GDP and approximately 40% of total exports.
- ▶ Equally, Moldovan research institutions acquired solid knowledge and experience, which further contributes to the development of agriculture.

The *Renewable Energy Assessment* performed by CRES in Moldova also showed increased potential for growing energy crops (table shown below). In the long term, sowing such crops could contribute to a lower dependence on the imported energy (which currently represents ~95% of total consumption).

Energy crops potential for biofuels (1st & 2nd generation) and bioenergy in Moldova (2008).

Crop	End use	Energy potential (PJ)
Oil crops	1 st gen Biodiesel	0.65
Sugar beet	1 st gen Bioethanol	2.47
Wheat	1 st gen Bioethanol	0.55
Maize	1 st gen Bioethanol	0.63
Perennial grasses	2 nd gen Bioethanol	3.10
	Heat & Electricity	6.12
Sweet sorghum	1 st gen Bioethanol	40.9

Source: CRES, Management calculations



- ▶ Law on Science and Technology Parks and Innovation Incubators no. 138-XVI / 21.07.2007 grants the following incentives to the residents of a science and technology park:
 - ✓ Exemption from payment of VAT (20%) on imported goods and services for 5 years from start of project;
 - ✓ Exemption from payment of VAT (20%) on goods and services bought on the territory of Moldova for 5 years from start of project;
 - ✓ Exemption from payment of customs taxes (5%) on imported goods and services for 5 years from start of project;
 - ✓ Exemption from payment of income tax (12%) during three tax periods, for 3 years from start of operations.

Being the first resident of the “Academica” Science and Technology Park , Tarus International can take advantage of the abovementioned legislative incentives.

- ▶ In conformity with EU Council Regulation (EC) No 55/2008 of 21 January 2008 introducing autonomous trade preferences for Moldova, bioethanol shall be admitted for import into the Community without quantitative restrictions or measures having equivalent effect and with exemption from customs duties and charges having equivalent effect;
- ▶ According to EU objectives called «20-20-20» the following developments should arise by 2020 :
 - ✓ 20% reduction of greenhouse gas emissions
 - ✓ Reduction in the use of fluid fuels by 20%
 - ✓ Increase in the use of renewable energy sources by 20%

Biofuels at a glance

Biofuels are a viable solution in addressing today's energy security concerns



- ▶ The oil crisis of the 1970s prompted interest in biofuels as an alternative to fossil fuels for use in transportation in many countries. However, subsequent drops in oil prices removed much of the incentives and stalled the momentum to expand biofuels production in most countries, with the notable exception of Brazil.
- ▶ Lately, issues related to energy supply security, oil price volatility and climate change mitigation caused a resurgence of interest in biofuels, with rapid expansion in output, mandates and targets to guarantee consumption, and investment in the development of advanced biofuel technologies.

Declining production costs are making biofuels more competitive, especially when oil prices are high, but in almost all cases, they still require subsidies to compete with gasoline and diesel in today's environment.



- ▶ Based on the feedstock used for production and the technologies used to convert that feedstock into fuel, biofuel technologies are classified in two groups:
 - ✓ **Conventional biofuels:** Technologies that normally utilize the sugar or starch portion of plants (e.g., sugarcane, sugar beet, cereals and cassava) as feedstock to produce ethanol and those utilizing oilseed crops (e.g., rape seed, sunflower, soybean and palm oil) to produce biodiesel are known as conventional (first generation) biofuels;
 - ✓ **Advanced biofuels:** biofuels produced using technologies that convert lignocellulosic biomass (e.g., agricultural and forest residues) are called advanced (second generation) biofuels, as are those produced from advanced feedstock (e.g., jatropha and micro-algae). As per The World Bank's World Energy Outlook, **advanced biofuels are assumed to enter the market by around 2020, mostly in OECD countries.**
- ▶ **Climate change consciousness has served as an important additional driver to the popularity of biofuels because it assists climate change mitigation efforts by displacing fossil fuel consumption. Given the enormous share of transportation in energy consumption, biofuels could contribute to the reduction of CO₂ emissions in the transport sector.**

Context and Regulation

- ▶ The EU is heavily dependent on fossil oil and fuel imports especially in the transport sector. In order to boost domestic ethanol production (and thus reduce dependency on imported energy), the EU adopted the Renewable Energy Directive, which included a 10% binding target for the use of renewable energy in road transport fuels by 2020.
- ▶ In order to comply with the EU directives, several member states have increased their national biofuels blending quotas for 2011 (please refer to Annex 4 for details).

Demand side

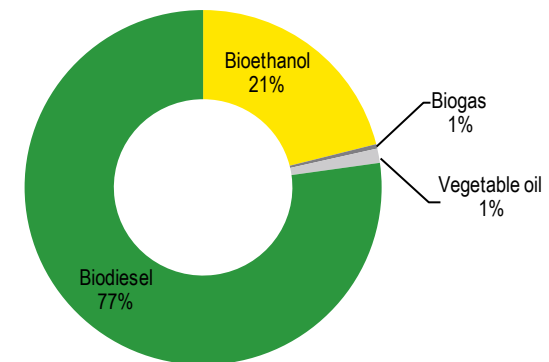
- ▶ As per IEA's World Energy Outlook, the global demand of biofuels is likely to double to around 220 billion liters over the next decade, leading to supply shortage of around 32 bn liters by 2020.
- ▶ The European consumption of bioethanol continued its exponential increasing trend, with a growth of 26.1% in 2011, compared to the year-ago figures. One of the main drivers of this growth was the gradual introduction of E10¹ in a number of EU countries.

Supply side

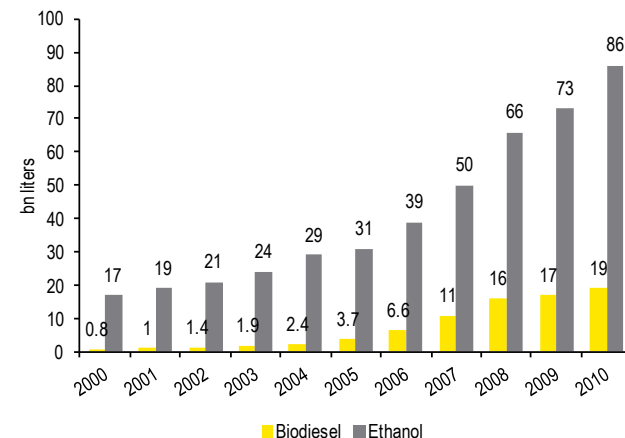
- ▶ The tail-off registered by Brazilian imports of bio-ethanol during the last year (mainly due to Brazil finding it hard to meet the demand of its own domestic flex-fuel vehicle market for bioethanol) should enable European production to grow as European consumption rises.
- ▶ Currently, most of the demand centers are far away from the supply centers, which increases distribution costs.

1. E10: Fuel mixture that contains 10% ethanol. Also known as gasohol

Breakdown of EU2010 biofuel consumption, by biofuel type (%)



Global production split between biodiesel and bioethanol



Source: Eurostat, EurObserver

Biofuels market overview (2/3)



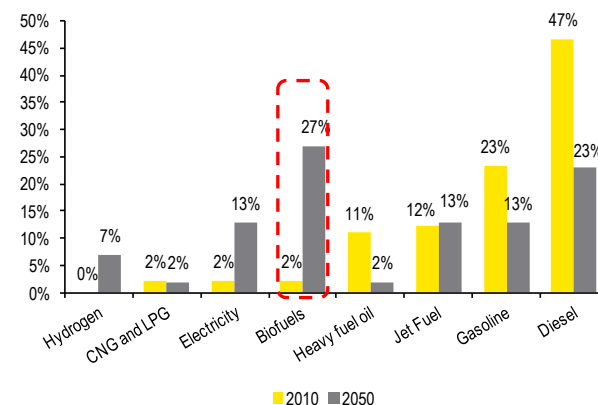
- The production capacities (as at Sept 2011) of the main European players in the bioethanol segment are detailed below (capacities shown in mn liters):

Company	Location of the production facility	No. of plants	Production capacity (mn. liters)	Main raw materials used
Abengoa	Spain (4)	14	3,180	Sugar cane, barley, wheat, cereals, raw alcohol, maize, lignocellulose
	United States (6)			
	Brazil (2)			
	France (1)			
	Netherlands (1)			
Tereos	France (6)	14	1,700	Sugar juice, wheat, sugar cane
	Brazil (6)			
	Belgium (1)			
	Czech Rep. (1)			
Crop Energies	Germany (1)	3	760	Sugar juice, cereals, wheat
	France (1)			
	Belgium (1)			
Cristanol	France(4)	4	540	Sugar juice, sugar beet, wheat, glucose, raw alcohol
Agrana	Austria (1)	2	410	Wheat, maize
	Hungary (1)			
Ensus	UK (1)	1	400	Wheat
Verbio	Germany (2)	2	300	Sugar juice, cereals

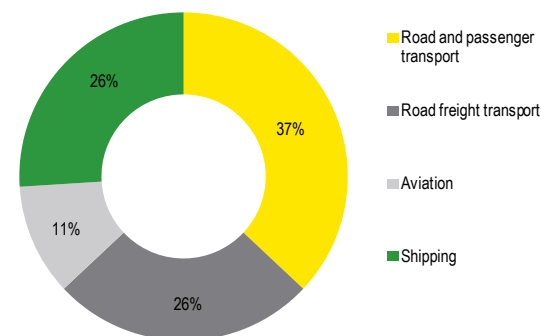
Future trends

- Based on the IEA's projections (please refer to Annex 3 for details), the worldwide biofuel demand will increase to 760 mn tonnes of oil equivalent (Mtoe), by 2050, up from ~110 mn tonnes in 2010.
- As per IEA's forecast, the increasing consumption calls for cumulative investment in biofuels production capacity of USD 335 bn. **More than half of this investment (i.e. USD 180 bn) is for conventional production of ethanol**, 10% for conventional biodiesel and only the remainder for advanced biofuels.

IEA forecast of global energy use in the transport sector, 2050



IEA forecast of the biofuels used by various transport models, 2050



Source: IEA, 2011c

Ethanol trading within the EU

- ▶ A significant share (~25%) of the European demand for ethanol is satisfied by imports from third countries. 2008 was a record year in this respect, total imports of ethanol (fuel and non-fuel) reaching at least 1 billion liters.
- ▶ European imports during 10m11 totaled 408 mn liters, similar to the level recorded during the same period of 2010 (415 mn liters).
- ▶ Ethanol is traded freely on a global market with Brazil as main trading partner (graphic on the right).

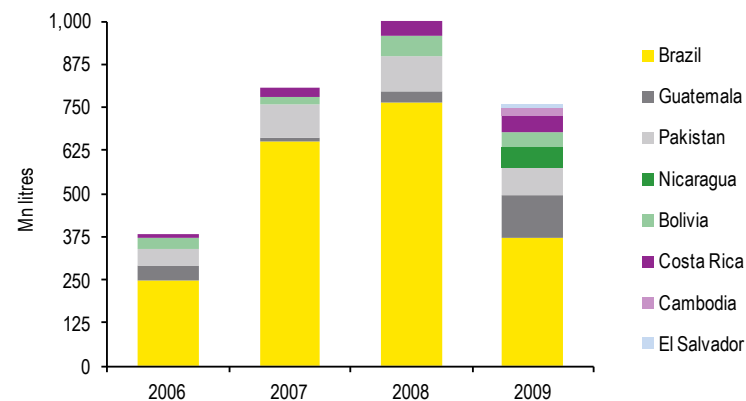
Ethanol production practices in Europe

- ▶ EU bioethanol is generally produced using a combination of sugar beets and wheat (bottom right table). **One of the main triggers for this development is the make-up of Europe's agricultural sector, which doesn't allow for massive plantations.** As a result, the European ethanol industry must draw upon a greater variety of feedstocks.
- ▶ The result is that **the European ethanol industry is fragmented** and unable to take advantage of a central feedstock supply network.
- ▶ Solutions for the growth of the industry may be offered by (1) more efficient production processes or (2) **innovative feedstock solutions, such as the one proposed by Tarus Intl.**

The raw material to be used by Tarus International is a completely new solution for the European market.

The sweet sorghum hybrids demonstrate higher productivity rates than grains: between 4.4 and 7.7 tonnes of ethanol can be obtained per ha, compared to 1.5-2.0 tonnes for other grains (Please refer to Section 3 for further information).

Breakdown of European imports of Ethanol, per trading partner



Source: IEA 2010

Raw materials used in various countries for the production of ethanol

Country	Ethanol feedstocks used (%)	Ethanol yields (l/hectare)
Brazil	Sugar cane (100%)	6,641
USA	Corn (98%)	2,770
	Sorghum seeds (2%)	1,365
China	Corn (70%)	2,011
	Wheat (30%)	1,730
EU-27	Wheat (48%)	1,702
	Sugar beet (29%)	5,145
Canada	Corn (70%)	3,460
	Wheat (30%)	1,075

Source: IEA, 2010

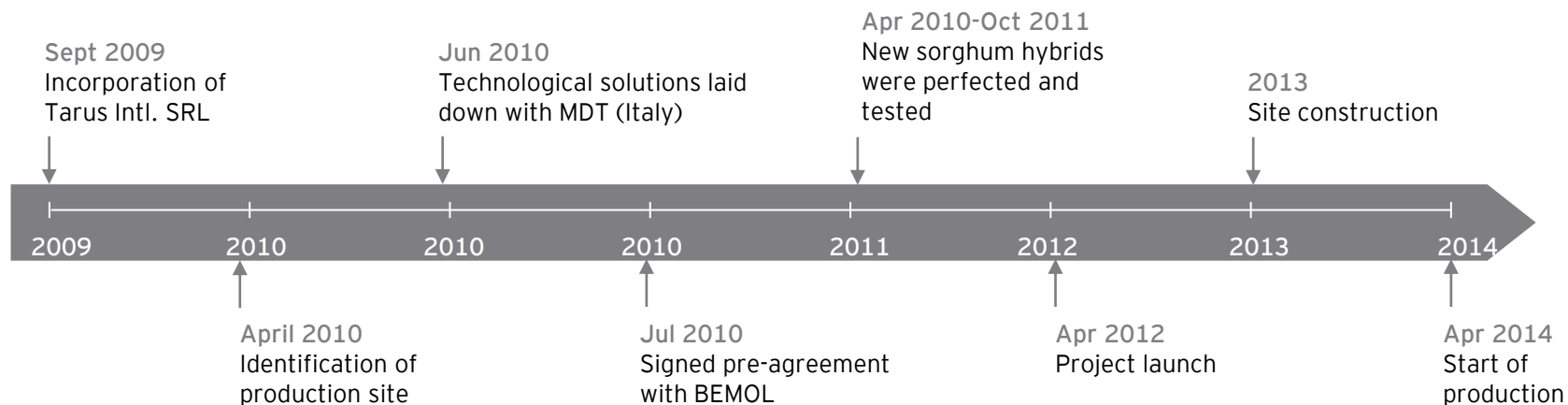
Business overview



Overview of Tarus International



Key stages of development



Tarus biorefinery produces more than just ethanol...

282,620 tonnes of sweet sorghum

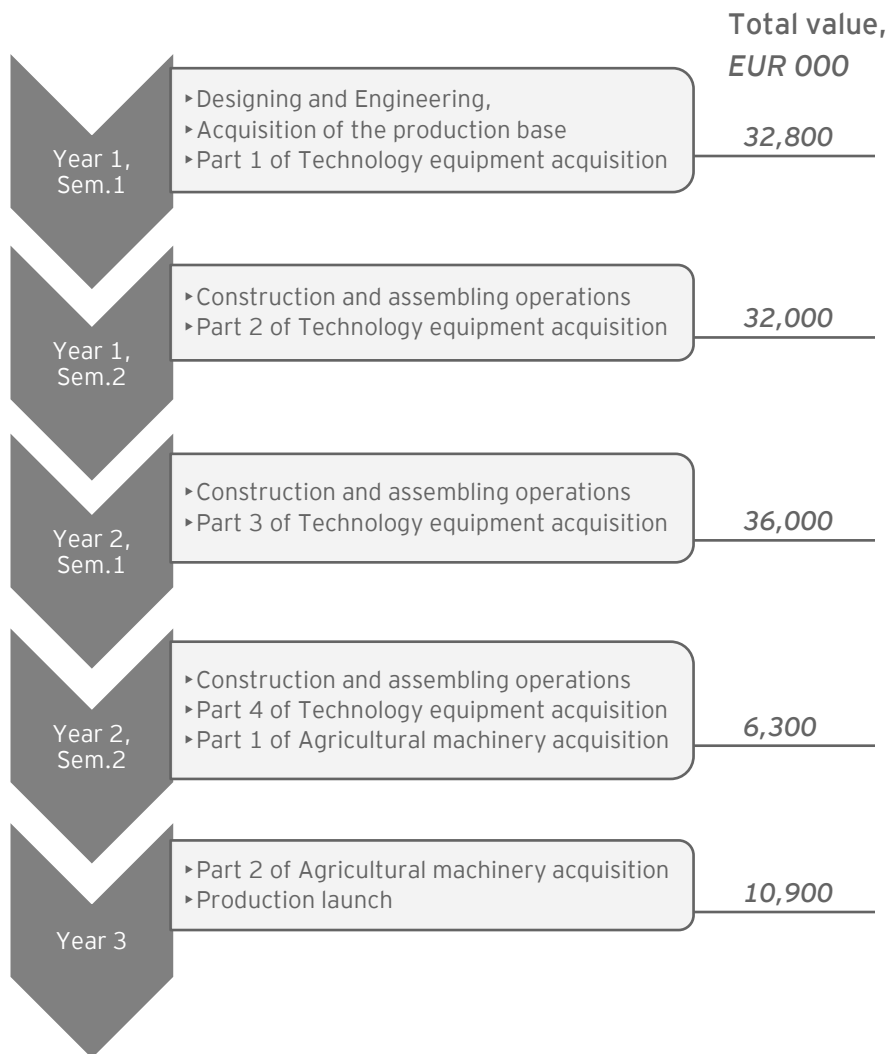
- 19.5 mn liters of bioethanol/year
- 24,960 MW h of electric energy/year

212,500 tonnes of corn seeds

- 87 mn liters of bioethanol/year
- 85,000 tonnes of CO₂/year
- 63,750 tonnes of DDGS/year

An ambitious project with highly experienced management and unique competitive advantages

Investment needs



Source: Management of Tarus Intl.

Investment split by categories

Investment area	EUR 000
Acquisition of the production base	7,500
Designing and Engineering	500
Construction and assembling operations	3,500
Technology equipment acquisition	96,500
Agricultural machinery acquisition	10,000
Total investment in equipments	118,000
Working capital needs	13,000
Total	131,000



Section 3

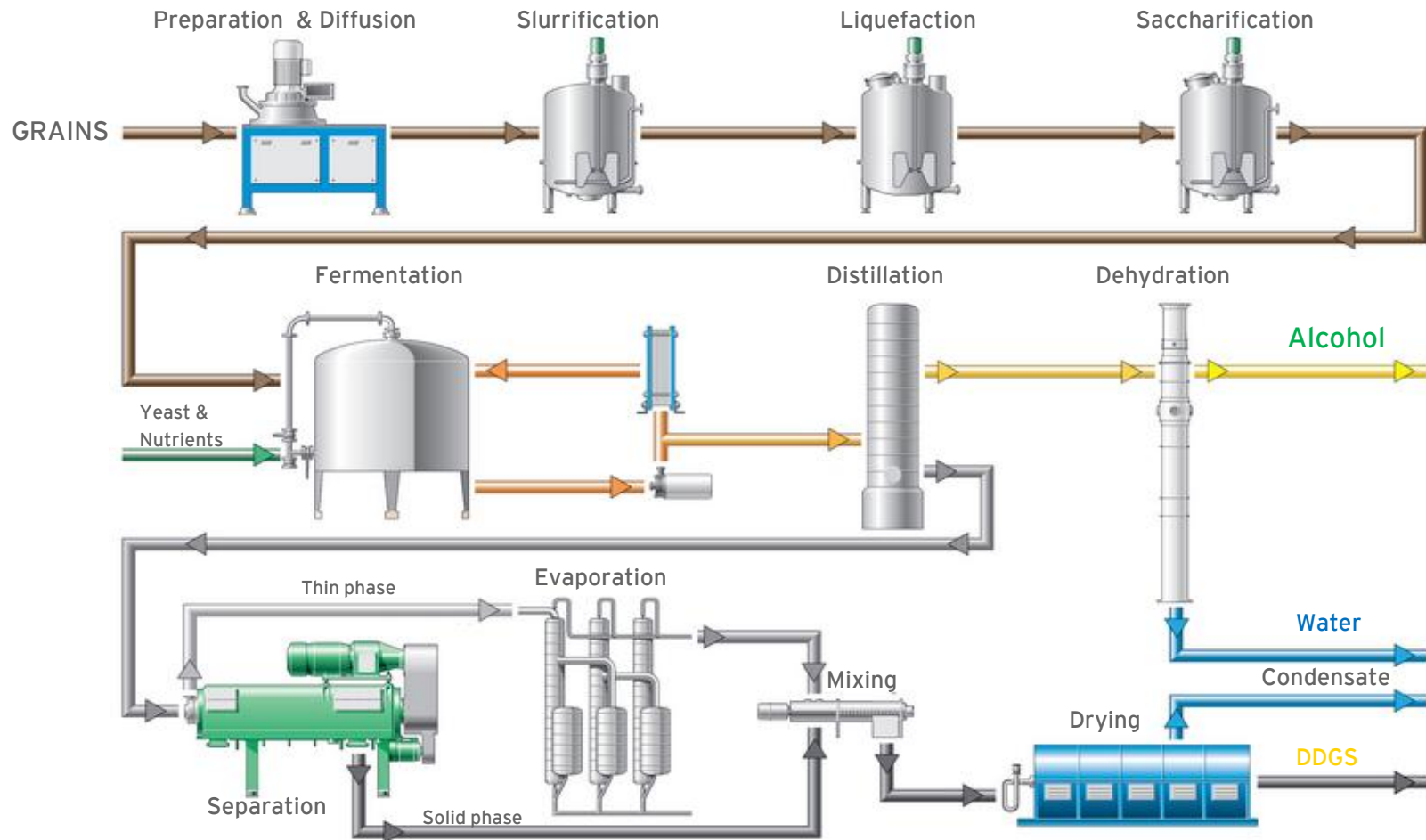
Products and operations



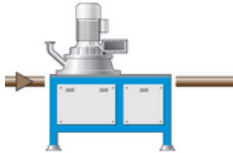
Production process and technology overview (1/4)



- The production process is depicted and explained in detail below:



Preparation & Diffusion



Once the raw materials reach the plant, they enter the preparation and diffusion process (dry milling). As compared to the traditional milling process, the advantages of the diffuser process are:

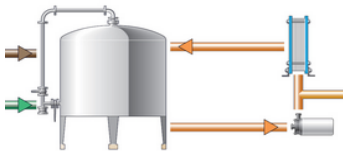
- ▶ lower investment cost;
- ▶ high extraction efficiency;
- ▶ better juice quality;
- ▶ lower maintenance cost.

Liquefaction & Saccharification



The next stage (liquefaction and saccharification²) of the production process results, by effect of added yeast and enzymes, in the reduction of the mash viscosity and in the breakdown of maltose molecules into glucose molecules ready for fermentation.

Fermentation



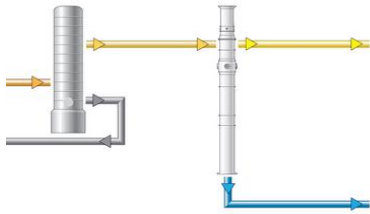
The fermentation unit is composed of 6 tanks (for both aerobic and anaerobic processes). **The production of ethanol can be improved by simultaneous fermentation** and saccharification process, whilst sugar-based feed-stocks contain simple sugars that can be extracted and fermented readily.

A series of centrifuges with vertical axis recover the yeast used during fermentation. After a regeneration process, the yeast is then reintroduced in the fermentation unit, thus improving the efficiency of this process. This process optimization translates into cost savings and better quality products.

During fermentation, alcohol and carbon dioxide are almost equally produced. This allows for the recovery of CO₂ which could be sold to the food and beverage industries.

2. The process of converting complex carbohydrate (e.g. starch) into simple monosaccharide components (e.g. glucose), through hydrolysis

Distillation & Dehydration

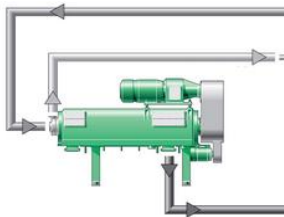


The alcohol mixtures obtained after fermentation have complex and among them different chemical compositions according to their raw materials sources.

Distillation and Dehydration are essentially the processes whereby the alcohol mixtures are separated obtaining the desired ethanol concentration and purity degree.

In order to process the 300,000 liters produced each day by the fermentation unit, the distillation equipment is composed of 3 cylinders. The result of this process is the hydrated ethyl alcohol with a concentration of 96.5%. **Further purity can be attained through dehydration, which removes remaining traces of water to produce anhydrous bioethanol with 99.8% purity, in line with the international standards for bioethanol obtained from biomass.**

Separation



The separation process is mainly applied for handling the stillage resulted from the whole process described above. Based on specific gravity difference, the process allows stillage suspended solid separation (GRITS) from the liquid fraction (thin stillage).

The mixture of these two products, previously concentrated and dried, result (after Mixing and Drying) in DDGS, that could be sold separately.

In order to avoid the appearance of organic waste, the biomass resulted from the production process is burned in specially-designed boilers, that allow both the burning of sweet sorghum biomass and the biogas (obtained in anaerobic tanks), resulted during the corn-processing period. The resulting steam powers two turbo-generator groups:

- ▶ Back-pressure based, producing **8 MW** of electric energy and a steam flow with the pressure of 2.5 bar, which is then distributed to the whole range of equipments included in the technological process.
- ▶ Condensation-based, producing an additional **16 MW** of electric energy.

- ▶ In order to facilitate the autonomous activity of the Company, six silos with a total surface of 24,000 sq m can store the corn seeds needed for 180 days of production.
- ▶ The plant will also be equipped with:
 - ▶ Tanks with a storing capacity of 13,5 mn liters of ethanol (45 days of production)
 - ▶ A mechanical division to provide any technical support and spare parts
 - ▶ An electrical energy transformation and connection plant, that will transfer the produced electrical energy surplus to the national grid
 - ▶ Administration offices
 - ▶ R&D and quality management laboratories
 - ▶ A computerized system that overlooks the whole production process
- ▶ The equipment will be provided by the Italian company MDT, one of the leading companies in the renewable energy technologies.



- ▶ Tarus is the first European bioethanol producer to use a highly efficient sorghum hybrid as raw material.
- ▶ Some of the main characteristics of the sorghum hybrid are:
 - ▶ Between 80 (in drought and non-irrigation conditions) and 140t (with irrigation) of biomass can be obtained from 1 sorghum-sowed hectare (as compared to max. 70 t/ha for sugar cane);
 - ▶ By comparison with other crops used as raw materials in this industry, the sorghum hybrid is up to 3-4 times more productive in terms of tonnes of ethanol extracted per ha (4.4-7.7 t/ha as compared to 1.5-2.0 t/ha for grains);
 - ▶ One hectare can absorb 55 tonnes of CO₂/year (compared to only 3-10 tonnes for grains). Payment from the Ecology Fund (financed by the World Bank) for the reduction of carbon dioxide emissions into the atmosphere is 10 €/t;
 - ▶ Sorghum is not demanding on soil fertility, bears a high degree of soil salinization and improves soil fertility by providing positive humus balance 1.5-2 t/ha/year. For other plants, this indicator is negative, ranging between 1.1t /ha/year of humus loss for wheat and 3.3t /ha/year loss for corn.

Further details on the characteristics of the sweet sorghum can be found in Annex 3.

Productivity of the main raw materials used for bioethanol production

Raw material	Crop yield (t/ha)	Bioethanol yield (t/ha)
Sugar cane	70	4.9
Cassava	40	6
Sweet sorghum	35	2.8
Corn	5	2.05
Wheat	4	1.56
Sweet sorghum - Tarus hybrid	80-140	4.4-7.7

Source: Balat & Balat (2008), Tarus Management

Bioethanol production cost estimates (exclusive of taxes), US cents per liter

Raw material	2012 cost	Forecasted cost (2030)
Bioethanol from sugar cane	25-50	25-35
Bioethanol from corn	60-80	35-55
Bioethanol from beet	60-80	40-60
Bioethanol from wheat	70-95	45-65
Bioethanol from lignocellulose	80-110	25-65
Bioethanol from Tarus hybrid	32-40	21-28

Source: Balat & Balat (2008), Tarus Management

Economics of bioethanol (1/3)

Cost elements for bioethanol production

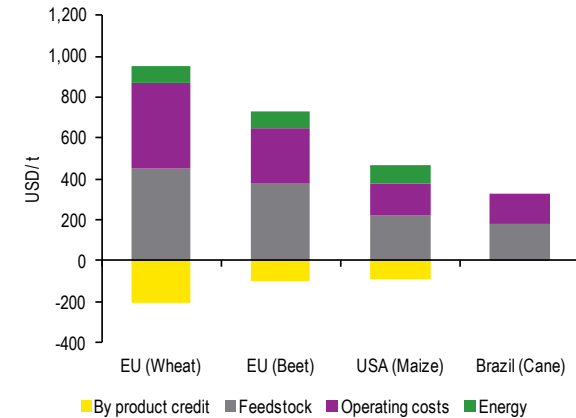


- ▶ The cost of supply³ of bioethanol varies widely depending on the technology, feedstock used and the size of the conversion plant (the total cost falls as the production capacity increases).
- ▶ Ethanol markets still have a regional structure.
- ▶ As shown in the graph on the right, two of the most important cost elements for bio-ethanol production are:
 - ▶ **The cost of feedstock (raw materials)**, which accounts for around 45-50% of the cost of supply. The choice of feedstock also explains, to a large extent, cost variation across countries;
 - ▶ **Energy and labour costs** which account for a further 30% of the total production costs.

Significant cost differences occur between regions, due to variations in (1) input (feedstock and land) costs, (2) labor costs, (3) transport costs and (4) processing time. As a secondary effect of such competitive advantages, the global trade of biofuels is also expected to increase.

- ▶ As per Management calculations, transportation costs can add some EUR 0.0020/liter/1000 km to the retail price (marine transport was considered).
- ▶ The IEA (Energy Technology Essentials - Biofuel Production) foresees a reduction of one-third in the cost of ethanol by 2030 due to technological improvements and lower costs of feedstock.

**Comparison of bio-ethanol production costs
(predominant raw material was considered for each region)**



Source: OECD, 2006

3. Ethanol energy content by volume is two-thirds that of gasoline, so costs refer to liter of gasoline equivalent (lge)

Economics of bioethanol (2/3)

Cost scenarios for biofuels compared to gasoline



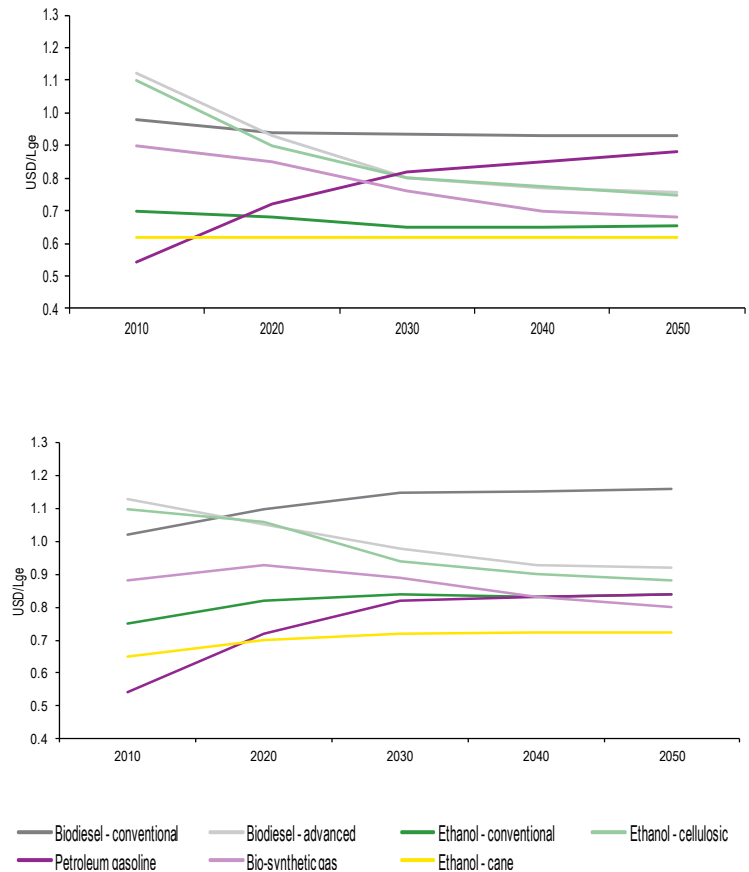
- ▶ While biofuels are expected to become increasingly competitive as fossil fuel prices rise and technologies mature, the strength of government support is a critical factor in determining how quickly this occurs.
- ▶ Regardless of the price scenarios considered by the IEA (graphs on the right), the price of crude oil is likely to increase exponentially by 2030, making the bioethanol price-competitive without further governmental subsidies. Please refer to Annex 1 for further details.
- ▶ Technological advances (higher crop yields, productivity increases) are likely to contribute to the cost reduction.

Most importantly, the co-generation of secondary products (DDGS, electrical energy recovery, CO₂) leads to subsequent costs recovery and improved overall financial performance.

The production process envisaged by the Tarus Intl. team will also generate:

- ▶ **24MW of electrical energy**, both for internal use (8MW) and external distribution (16 MW). The sale of renewable energy is guaranteed by the Renewable Energy Law in RM;
- ▶ Roughly one third of the cereals used for bioethanol is recovered in the production process and goes back to the food chain as protein-rich animal feed called **DDGS**;
- ▶ CO₂ that can be sold in the food and beverages industry.

**Blue Map Low-Cost scenario (Graph 1)
and High-Cost Scenario (Graph 2)**



Source: IEA, BLUE Map Scenario

Note: Costs reflect global average retail prices, without taxation. Regional differences can occur depending on feedstock prices and other cost factors

Economics of bioethanol (3/3)

Bioethanol pricing



Domestic prices of ethanol

- Prices have been subject to significant increases over the last few quarters, supported by tight supply and high feedstock costs. **On December the 2nd 2011, US corn-based ethanol hit \$2.89 a gallon**, the highest ever recorded for the month, according to data compiled by Bloomberg.
- Futures have climbed 11% this year on the Chicago Board of Trade because of shortfalls in key overseas markets.
- This situation contrasts with 2007/08 where ethanol price movements did not follow the pace of the commodity price increases and ethanol profit margins were reduced.

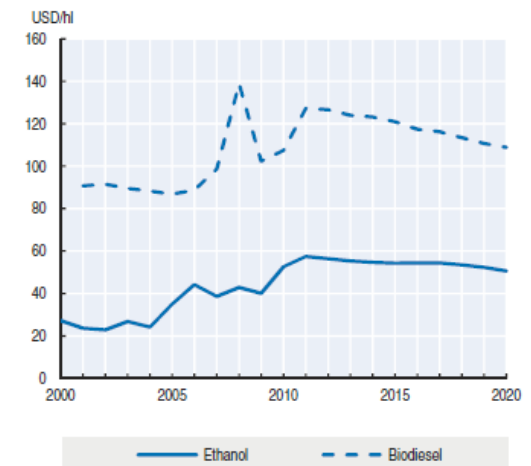
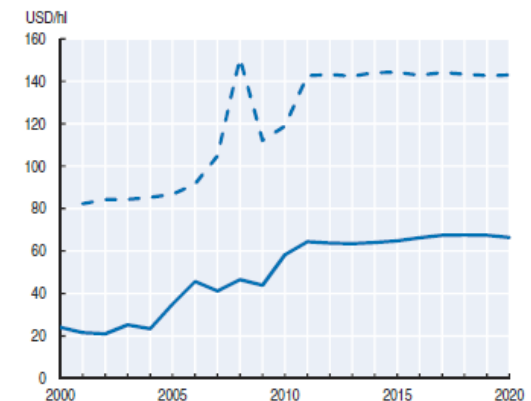
Ethanol category/Region	Ccy/u.m.	Current price range (as at Jul11)	Price range, one year ago (as at Jul10)
I. Beverage - 96% purity			
UK	€/hl	74-79	66-67
France	€/hl	72-78	62-64
Germany	€/hl	75-79	62-64
Italy	€/hl	80-84	62-64
II. Industrial - 99% purity			
UK	€/hl	101-113	89-94
France	€/hl	78-82	68-71
Germany	€/hl	79-84	68-72
Italy	€/hl	85-88	68-72

Sources: ICIS Pricing

Projection highlights

- Prices are projected to be on average 80% higher than over the previous decade in the case of ethanol and 45% in the case of biodiesel. As per OECD, they are projected to reach, respectively, USD 66.4 per hl and USD 142.9 per hl by 2020.
- Prices should decrease slightly when expressed in real terms over the outlook period but the ratios of biofuel prices to major biofuel feedstock prices are expected to remain relatively stable (please refer to graphs on the right for details).

Evolution of prices expressed in nominal terms (Graph 1) and in real terms (Graph 2)



Sources: OECD Agricultural Outlook 2011



Section 4

Marketing and sales strategy



Section 4

Marketing and sales strategy

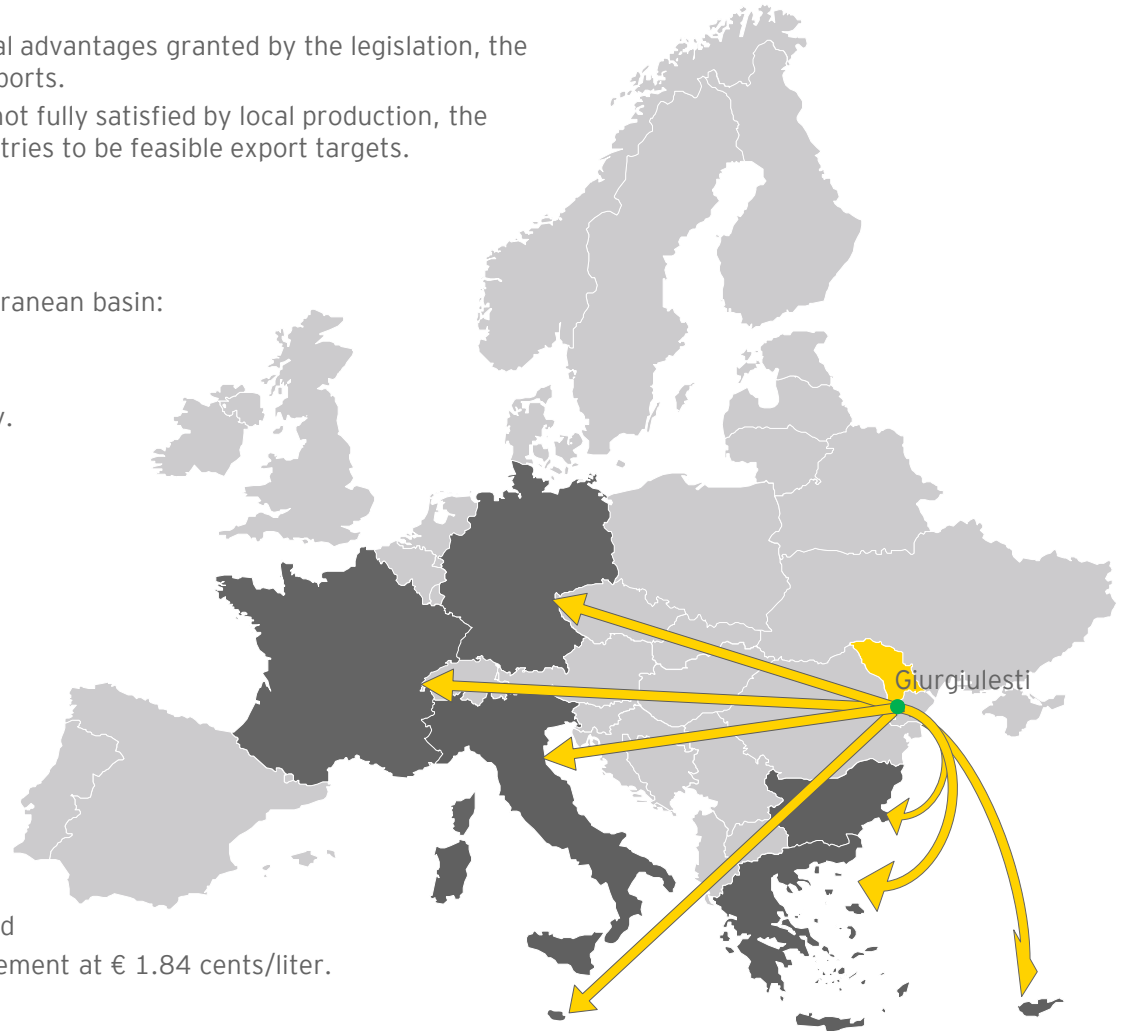
- ▶ Due to the small domestic market and the fiscal advantages granted by the legislation, the sales strategy of the company will focus on exports.
- ▶ Since the European demand for bioethanol is not fully satisfied by local production, the Company's Management considers all EU countries to be feasible export targets.
- ▶ **A strategic focus will be placed on:**
 - ▶ countries within the Black Sea's and Mediterranean basin:
 - ▶ Greece , Italy, France, Turkey.
 - ▶ Danubian countries:
 - ▶ Romania, Bulgaria, Hungary, Germany.

, which can be easily reached by sea through the Giurgiulesti port.

As per Management estimates, the transportation costs are estimated at:

- ▶ € 0.0026 per liter for Thessaloniki, Greece.
- ▶ € 0.0065 per liter for Ravenna, Italy.

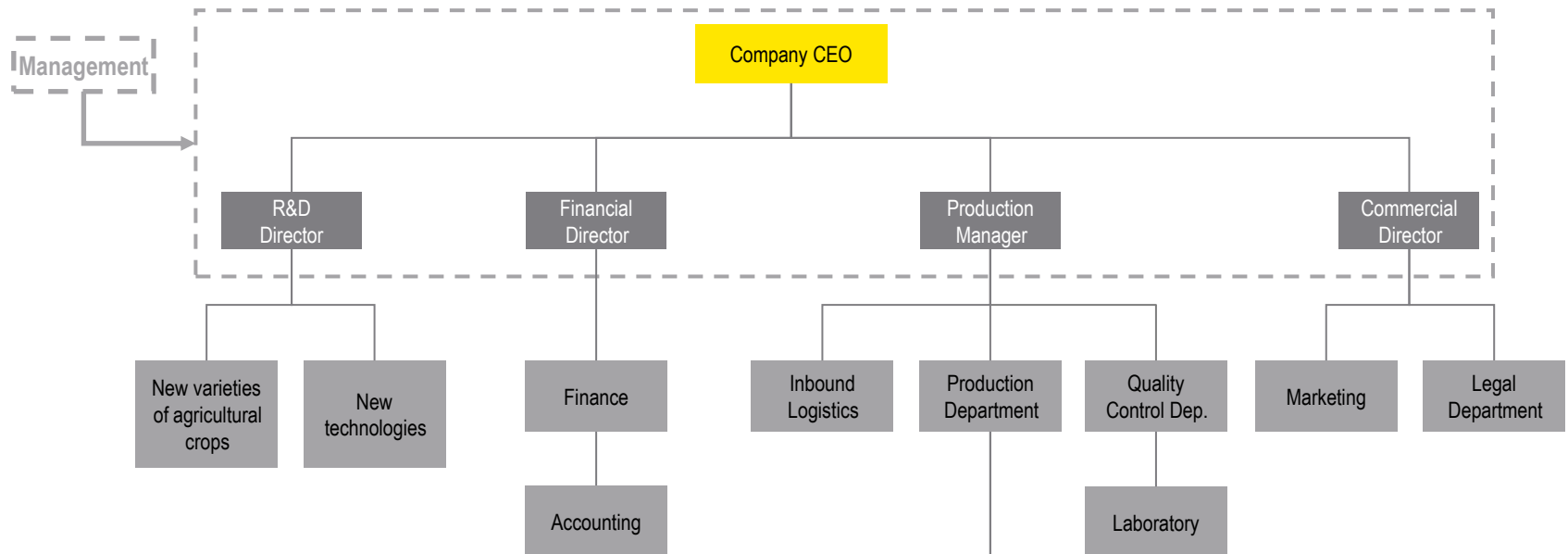
Also, the cost of transition through both Bosfor and Dardanele narrow straits was estimated by Management at € 1.84 cents/liter.



Human Resources



Organizational chart



Employees by department

Administrative personnel	22
Commercial department	19
Production department	
Bioethanol production plant	62
Production of agricultural crops	74
Quality control department	5
Inbound Logistics	7
Accounting department	2
R&D department	9
TOTAL	200

Virgiliu Rusu



- ▶ 43 years old

Education

- ▶ Graduated from the Technical University of Moldova, the Technology Department, with a diploma of plant-operating engineer, winemaker.

His experience includes:

- ▶ GM of Subsidiary Enterprise "Crimean Wine House" Ltd. Feodosia, Crimea, Ukraine.
- ▶ Deputy director of Subsidiary Enterprise "Crimean Wine House" Ltd. "L'empire du vin" Feodosia, Crimea, Ukraine.
- ▶ Chief plant-operating engineer, winemaker at the "Setvin Group" Ltd Joint Venture winery.
- ▶ Deputy GM and Chief plant-operating engineer, winemaker at the "Dionisimus" JSC winery.

▶ President of the Moldovan Association of Renewable Energy Producers.

Contact:

- ▶ Email: virgiliu.r@biofuel-tarus.eu
- ▶ Tel: 0037368888080

Veaceslav Afanasiev



- ▶ 57 years old

Education

- ▶ Graduated as physicist, from Moscow State University "M. Lomonosov".
- ▶ PhD in Agriculture, Chisinau Agricultural Institute.

His experience includes:

- ▶ GM at "Enecom International" SRL.
- ▶ Deputy minister, Ministry of Economy.
- ▶ Deputy Director, Institute for Development of Information Society of Academy of Sciences of Moldova.
- ▶ Director of the Agency for Innovation and Technology Transfer, within the Academy of Sciences of Moldova.

▶ Membership of the following professional bodies:

- Russian Academy of Natural Sciences.
- International Academy of Ecology and Life Protection Sciences.
- International Academy of Agrarian Education.

Contact:

- ▶ Email: veaceslav.a@biofuel-tarus.eu
- ▶ Tel: 0037368612172

The background is a complex, hand-drawn collage of various icons and symbols. It includes mathematical elements like numbers (e.g., 1+1, 971, 321, 47), percentages, and geometric shapes. There are also depictions of everyday objects such as a television, a CD, a calculator, and a lightbulb. Nature-related motifs include flowers, leaves, and wind turbines. Social or community themes are represented by drawings of people holding hands and a group of children. The overall style is whimsical and illustrative, suggesting a broad range of topics or concepts.

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Business Plan assumptions (1/2)



- ▶ The business plan considers April 2012 as the start date of the forecasting period and is drafted for a period of 7 years, ending in December 2018.
 - ▶ The currency of the BP is EUR, as most of the cash-flows generated by the business are either in EUR or EUR indexed. Throughout the BP, computations are done in EUR, with the final results of the model presented in EUR'000.
 - ▶ The Financial Statements presented on the following slides take into account an inflation rate forecasted by NBM at 8.8% p.a.
 - ▶ As a result of the fiscal advantages detailed in the previous sections, no income tax has been accounted for in the first 3 years of operations.
-
- ▶ Most of the inputs concerning productivity rates have been provided by the Management of Tarus Intl.
 - ▶ Interest payable has been calculated based on a yearly cost of debt of 5.5%.

FEEDSTOCK INPUTS		
	u.m.	Quantity
1. Production capacity, maximum		
1.1 Corn	t/day	733
1.2 Sorghum	t/day	4,348
2. Agriculture land for crops		
2.1 Corn	ha	25,000
2.2 Sorghum	ha	3,533
3. Crops yield		
3.1 Corn	t/ha	8.5
3.2 Sorghum	t/ha	80
4. Feedstock – internally generated		
4.1 Corn	t	212,500
4.2 Sorghum	t	282,620

TRADING INPUTS		
	u.m.	Quantity
1. Bio-ethanol - liters produced as % from feedstock		
1.1 Corn	liters per t. of feedstock	410
1.2 Sorghum	liters per t. of feedstock	69
1.3 Unit price per liter	€	0.76
2. Electricity		
2.1 MWh per day produced and sold	MWh	384
2.2 Unit price per MWh	€	70
3. DDGS		
3.1 Quantity	as % from corn	30%
3.2 Unit price per t.	€	167
4. CO2		
4.1 Quantity	as % from corn	40%
4.2 Unit price per t.	€	5

Business Plan assumptions (2/2)



- ▶ Number of employees will increase gradually to 200 employees in 2014.
- ▶ Sweet sorghum-related costs are accounted for beginning with Jan 2014, when the production is scheduled to start.
- ▶ Corn production-related costs are scheduled to start in 2013, as Tarus plans to build raw materials reserves before the start of production.

COST OF SALES INPUTS				
	u.m.	2012	2013	2014 -
1. Raw materials cost (feedstock internally generated)				
1.1 Corn (seed, fertilization, herbicide, other variable and fixed costs)	€/t	-	25.5	25.5
1.2 Sorghum (seed, fertilization, herbicide)	€/t	-	-	4.8
2. Electricity cost				
3.1 During the period of corn seeds processing	MWh per day	-	-	192
3.2 Unit price per MWh	€	-	-	96
3. Personnel cost				
4.1 No. of employees	no.	80	120	200
4.2 Average monthly salary	€	500	500	500
4.3 Social security contributions	% of gross salary	26%	26%	26%
4. Fuel cost				
4.1 Corn	l/ha	-	70	70
4.2 Sorghum	l/ha	-	-	90
4.3. Unit price per liter	€	1	1	1
5. Land lease cost				
5.1 Corn	€/ha/year	-	80	80
5.2 Sorghum	€/ha/year	-	80	80

- ▶ Maintenance of agricultural machinery is scheduled to increase after two years of operations, to approx. EUR 0.5 mn/year starting with 2016.
- ▶ The budget allowance for the R&D department was estimated at EUR 0.2 mn per year.
- ▶ Maintenance of processing and production lines was estimated at EUR 1 mn per year.

OTHER COSTING INPUTS				
	u.m.	2012 - 2013	2014 - 2015	2016 -
1. Maintenance cost inputs (annual costs)				
1.1 Agricultural machinery	€	-	200,000	450,000
1.2 Processing and production lines	€	-	1,000,000	1,000,000
2. R&D (annual costs)				
2.1 Materials and other expenses	€	250,000	250,000	250,000
3. Insurance costs				
3.1 Sweet sorghum insurance cost	€/ha	-	75	75
3.2 Corn insurance cost	€/ha	68	68	68
4. Working capital days				
4.1 Debtor days	days	30	30	30
4.2 Creditor days	days	30	30	30
5. CAPEX inputs				
5.1 Straight line depreciation % for acquisition of assets from existing company	%	20		
5.2 Straight line depreciation % for agricultural machinery	%	8		
5.3 Straight line depreciation % for CAPEX	%	10		

Financial summary (1/3)

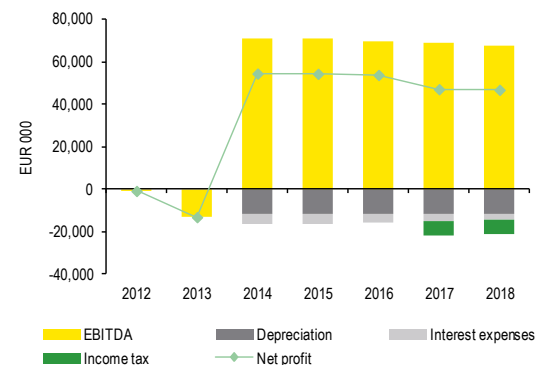


Key Financials – Profit & Loss

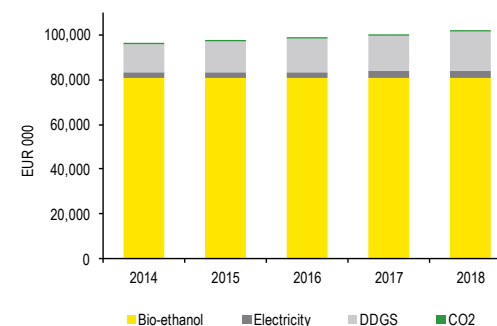
Currency: € 000

	01 Apr 12	01 Apr 13	01 Apr 14	01 Apr 15	01 Apr 16	01 Apr 17	01 Apr 18
	31 Mar 13	31 Mar 14	31 Mar 15	31 Mar 16	31 Mar 17	31 Mar 18	31 Mar 19
Sales	-	-	96,209	97,545	98,997	100,578	102,298
Operating expenses							
Raw materials	-	(5,896)	(8,020)	(8,726)	(9,494)	(10,329)	(11,238)
Electricity	-	-	(6,327)	(6,884)	(7,490)	(8,149)	(8,866)
Salaries	(605)	(987)	(1,790)	(1,947)	(2,119)	(2,305)	(2,508)
Fuel	-	(1,904)	(2,448)	(2,663)	(2,898)	(3,153)	(3,430)
Land lease	-	(2,283)	(2,283)	(2,283)	(2,283)	(2,283)	(2,283)
Maintenance cost	-	-	(1,420)	(1,545)	(2,032)	(2,211)	(2,405)
R&D	(250)	(272)	(296)	(322)	(350)	(381)	(415)
Insurance cost	-	(1,855)	(2,331)	(2,536)	(2,759)	(3,002)	(3,266)
Gross profit / EBITDA	(855)	(13,196)	71,294	70,638	69,573	68,766	67,887
Gross profit margin (%)			74%	72%	70%	68%	66%
Depreciation	-	-	(12,056)	(12,056)	(12,056)	(12,056)	(12,056)
Operating profit	(855)	(13,196)	59,238	58,581	57,517	56,709	55,830
Operating profit margin (%)			62%	60%	58%	56%	55%
Interest paid	-	-	(4,741)	(4,214)	(3,687)	(3,160)	(2,634)
Profit before tax	(855)	(13,196)	54,497	54,367	53,830	53,549	53,197
Income tax	-	-	-	-	-	(6,426)	(6,384)
Net profit	(855)	(13,196)	54,497	54,367	53,830	47,123	46,813
Net profit margin			57%	56%	54%	47%	46%
Retained profit control account							
Retained profit b/f	-	(855)	(14,051)	40,446	94,813	148,643	195,766
Net profit	(855)	(13,196)	54,497	54,367	53,830	47,123	46,813
Retained profit c/f	(855)	(14,051)	40,446	94,813	148,643	195,766	242,579

EBITDA to Net Profit derivation



Breakdown of revenue by type of product



- ▶ The breakdown of sales by type of product shows a 80/20 split between revenues generated by bioethanol and secondary products, respectively.

Financial summary (2/3)

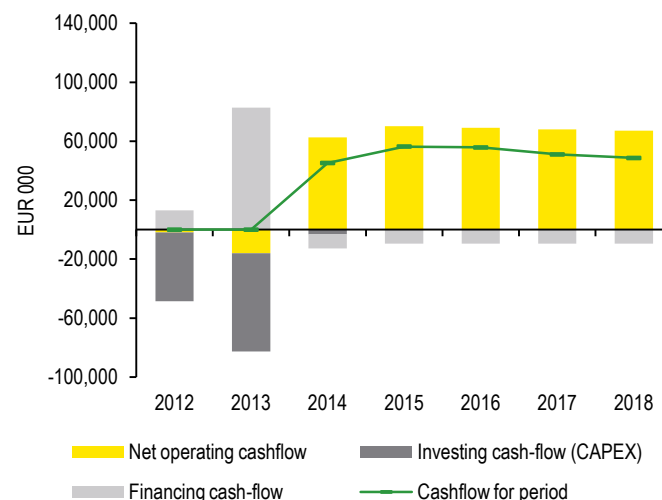


Key Financials – Cash flow

Currency: € 000

	01 Apr 12	01 Apr 13	01 Apr 14	01 Apr 15	01 Apr 16	01 Apr 17	01 Apr 18
	31 Mar 13	31 Mar 14	31 Mar 15	31 Mar 16	31 Mar 17	31 Mar 18	31 Mar 19
Operating profit	(855)	(13,196)	59,238	58,581	57,517	56,709	55,830
Depreciation	-	-	12,056	12,056	12,056	12,056	12,056
(Increase) / decrease in inventories	(1,098)	(3,586)	(1,688)	(542)	(629)	(664)	(722)
(Increase) / decrease in debtors	-	-	(7,908)	(88)	(141)	(451)	(170)
Increase / (decrease) in creditors	70	862	924	141	194	421	213
Net operating cash flow	(1,883)	(15,921)	62,623	70,150	68,997	68,071	67,208
Capital expenditure	(46,568)	(66,735)	(3,101)	-	-	-	-
CIT paid	-	-	-	-	-	(4,313)	(6,397)
Debt (repayments)/drawdowns	13,115	82,655	(9,577)	(9,577)	(9,577)	(9,577)	(9,577)
Net interest paid	-	-	(4,741)	(4,214)	(3,687)	(3,160)	(2,634)
Equity injected	35,336	-	-	-	-	-	-
Cash flow for period	-	-	45,204	56,359	55,733	51,021	48,600
Cash control account							
Balance b/f	-	-	-	45,204	101,563	157,295	208,316
Cash flow for period	-	-	45,204	56,359	55,733	51,021	48,600
Balance c/f	-	-	45,204	101,563	157,295	208,316	256,916

Period cash-flow split by nature



- ▶ The Cash flow statement has been prepared using the indirect method, i.e. starting from the net financial result of the period and adding back non-financial items and changes in the NWC items, in order to reach the Operating cash-flow.
- ▶ The cash-flow results are summarised in the chart on the right.

Financial summary (3/3)

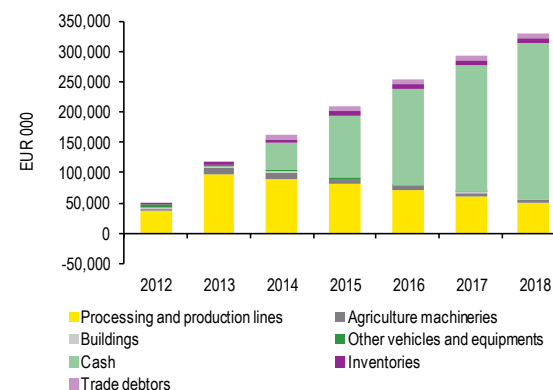


Key Financials – Balance sheet

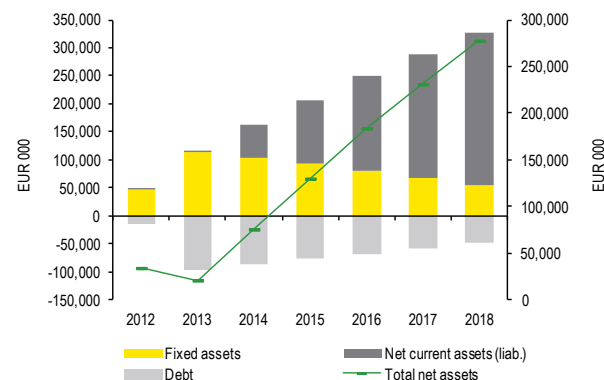
Currency: € 000	01 Apr 12	01 Apr 13	01 Apr 14	01 Apr 15	01 Apr 16	01 Apr 17	01 Apr 18
	31 Mar 13	31 Mar 14	31 Mar 15	31 Mar 16	31 Mar 17	31 Mar 18	31 Mar 19
Fixed assets	46,568	113,303	104,348	92,291	80,235	68,178	56,122
Cash	-	-	45,204	101,563	157,295	208,316	256,916
Inventories	1,098	4,684	6,372	6,914	7,543	8,207	8,929
Trade debtors	-	-	7,908	7,995	8,137	8,588	8,758
Total current assets	1,098	4,684	59,484	116,472	172,975	225,111	274,603
Trade creditors	(70)	(932)	(1,856)	(1,998)	(2,192)	(2,612)	(2,826)
CIT	-	-	-	-	-	(2,113)	(2,099)
Net current assets/(liabilities)	1,028	3,752	57,628	114,475	170,784	220,386	269,678
Debt	(13,115)	(95,770)	(86,193)	(76,616)	(67,039)	(57,462)	(47,885)
Net assets	34,481	21,285	75,782	130,149	183,979	231,102	277,915
Represented by:							
Share capital	35,336	35,336	35,336	35,336	35,336	35,336	35,336
Retained earnings	(855)	(14,051)	40,446	94,813	148,643	195,766	242,579
Total shareholders funds	34,481	21,285	75,782	130,149	183,979	231,102	277,915

- ▶ The main assumption regarding financing is that during the BP period the Investors will use the cash generated by the business to repay the existing loans and hence, refinance the Company without attracting new debt.
- ▶ Fixed assets concern the processing and production lines to be acquired by 2014, agricultural machinery, as well as other tangible assets - currently in the property of the farm that will be acquired.

Evolution of assets structure and value



Evolution of net assets



Investment analysis and financial ratios

Investment analysis and financial ratios

Based on the previously-mentioned Cash-flow statement, the investment's internal rate of return was computed at investor level, by using **Free Cash-Flows to Equity** (FCFE - i.e. net cash variance in each period, adjusted for the impact of any dividend distribution, plus the residual equity book value at the end of the BP). The results are presented in the table below:

Currency: € 000	U.M.	Indicator value
IRR (Equity level)	%	75.9
Minimum EBITDA margin	%	66

Source: EY computations

Financing ratios - These ratios are following the capital structure evolution and are split into:

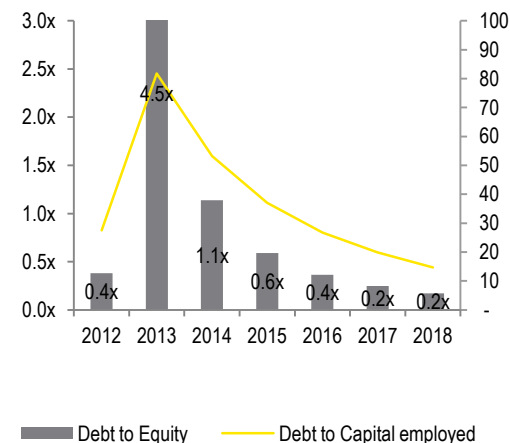
Overall leverage:

- ▶ Debt to Capital employed - steadily decreases as the initial loans are repaid during the period of the BP, from a high level of 82% in 2013 to 15% in 2018. Most of the debt will be used in 2013 (EUR 80 mn, both for payment of the production line and for working capital needs), causing the spike observed in the graph on the right.
- ▶ Debt to Equity - is a similar indicator, that sees an identical trend during the period of the BP, decreasing from 4.5 times the value of Equity in 2012, to 0.2 in 2018.

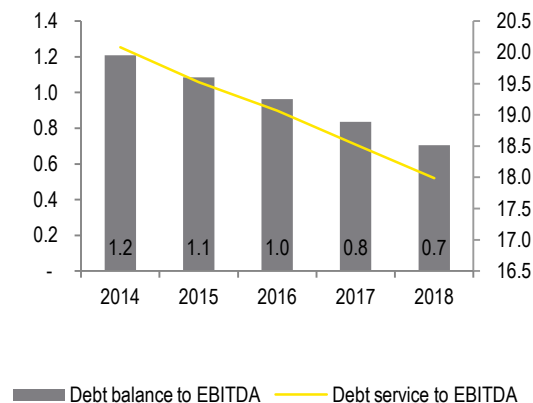
EBITDA-Debt coverage:

- ▶ Debt service to EBITDA - shows how much of the EBITDA goes towards repaying the company's debt in a certain period. The ratio has been computed starting with 2014, as that is the first year with principal repayment scheduled. This indicator is well below 100% in all the BP periods, indicating that forecasted EBITDA is high enough to support the estimated debt repayments.
- ▶ Debt balance to EBITDA - Debt is forecasted not to exceed the conservative level of 3x EBITDA and is constantly declining towards the end of the BP.

Leverage indicators



Debt coverage of EBITDA

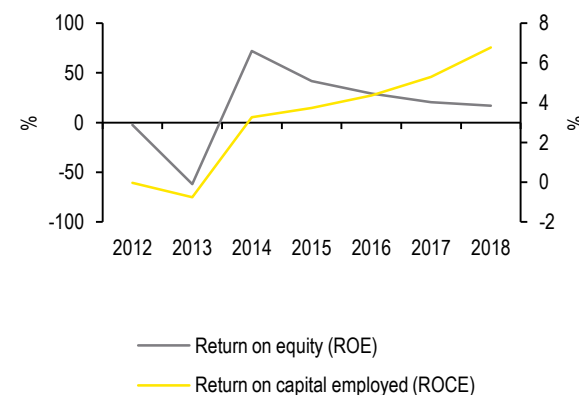


Profitability ratios:

The most relevant profitability ratios were considered to be :

- ▶ Return on capital employed (ROCE) - computed as EBIT over (Debt plus Equity). The indicator steadily increases from -0.03% in 2012 to 6.78% at the end of the BP.
- ▶ Return on equity (ROE) - computed as Net profit over Equity. The indicator see yearly decreases throughout the BP (from 72% in 2014 to 16.8% in 2018), as the accumulated retain earnings increase the Equity size and hence the base of the computation fraction. 2012 and 2013 sees a negative return on equity.

**ROCE and ROE
forecasted evolution**



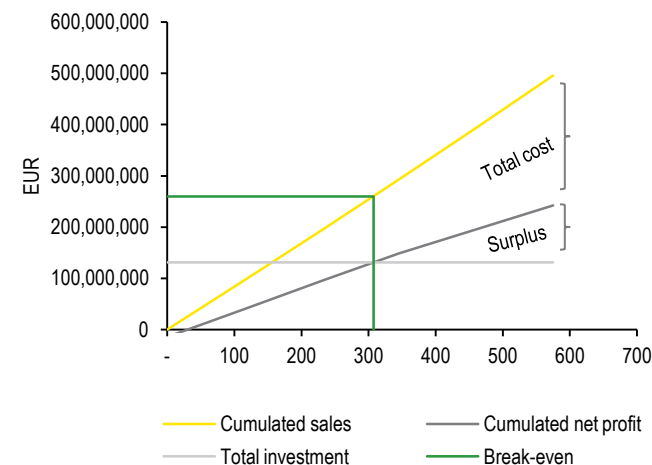
Currency: € 000		U.M.	2012	2013	2014	2015	2016	2017	2018
Leverage									
Debt to Capital employed	%		28	82	53	37	27	20	15
Debt to Equity	x times		0.4x	4.5x	1.1x	0.6x	0.4x	0.2x	0.2x
EBITDA coverage of Debt									
Debt service to EBITDA	%		-	-	20.1	19.5	19.1	18.5	18.0
Debt balance to EBITDA	x times		15.3	7.3	1.2	1.1	1.0	0.8	0.7
Profitability ratios									
Return on capital employed (ROCE)	%		(0.03)	(0.75)	3.26	3.74	4.36	5.29	6.78
Return on Equity (ROE)	%		(2.5)	(62.0)	71.9	41.8	29.3	20.4	16.8

Source: EY computations

Break-even analysis:

- ▶ In the break-even analysis it was determined the level of sales that are required to cover the initial investment of EUR 131 mn.
- ▶ As it is depicted in the right graph, Tarus need to produce and sell a minimum level of 307 mn liters of bioethanol in order to cover the initial investment. These 307 mn of bioethanol are estimated to be sold at a price of EUR 0.76 per liter.
- ▶ Based on the production capacity of 100 mn litres per year it is estimated that this level of sales will be achieved in the third year of production.

Break-even point graph



Currency: € 000

	2012	2013	2014	2015	2016	2017	2018
Cumulated net profit/(loss)	(855)	(14,051)	40,446	94,813	148,643	195,766	242,579
Total investment	131,106	131,106	131,106	131,106	131,106	131,106	131,106
Surplus/deficit	(131,961)	(145,157)	(90,660)	(36,293)	17,537	64,660	111,473



Appendix



- ▶ **New Policies Scenario** - Within the World Energy Outlook 2011, the IEA presents projections for three scenarios, the most comprehensive of them being The New Policies Scenario. The NPS is based on broad policy commitments and plans that have been announced by countries around the world to address energy security, climate change and local pollution, and other pressing energy-related challenges, even where the specific measures to implement these commitments have yet to be announced. More specifically, the New Policies Scenario takes into account all policies and measures included in the Current Policies Scenario (one of the other two scenarios), as well as the following:
 - ▶ Cautious implementation of recently announced commitments and plans, including the Cancun Agreements.
 - ▶ Continuation of an Emissions Trading Scheme (ETS) in Europe and New Zealand; introduction of CO₂ pricing through taxes/ETS in Australia from mid-2012, in Korea from 2015 and in China from 2020. Shadow price of carbon adopted from 2015 in Canada, Japan and the United States, influencing solely new investment decisions in power generation. Access to international offset credits is assumed for all countries participating in an ETS.
 - ▶ Phase-out of fossil-fuel consumption subsidies in all net-importing regions by 2020 (and, as in the Current Policies Scenario, in net-exporting regions where specific policies have already been introduced).
 - ▶ Extension by five years of the nuclear plant lifetimes assumed in the Current Policies Scenario, in those countries that have confirmed their intention to continue or expand the use of nuclear energy, accounting for about 50% of the plants currently in operation.
 - ▶ Extended support for renewables-based electricity-generation technologies and biofuels for transport.
 - ▶ Heavy-duty vehicle fuel efficiency standards in the United States; light-commercial vehicle and passenger light-duty vehicle (PLDV) emissions per kilometer targets in the European Union; PLDV fuel efficiency standards in India and China.
 - ▶ For 2020 to 2035, additional measures that maintain the pace of the global decline in carbon intensity - measured as emissions per dollar of gross domestic product, in purchasing power parity terms - established in the period 2009 to 2020.
- ▶ **BLUE Map Scenario** - Another IEA forecast, which assumes that GHG emissions will be reduced to half their 2005 levels by 2050, being also broadly optimistic for all other forecasted factors (investments made, decreasing demand for petroleum products, etc).

Source: IEA World Energy Outlook

http://www.iea.org/weo/docs/weo2011/other/WEO_methodology/WEO2011_AnnexB.pdf

<http://www.iea.org/work/2011/egrd/day1/Remme.pdf>

Appendix 2

Biofuels blending quotas in the EU



Country	TOTAL % *	Biodiesel %*	Bioethanol %*
Austria	5.75	min. 6.3	min. 3.4
Belgium	4 (vol.)	n.a.	n.a.
Bulgaria	5 (vol.)	4 (vol.)	2 (vol.)
Cyprus	2.5	n.a.	n.a.
Czech republic	n.a.	6 (vol.)	4.1 (vol.)
Denmark	3.5	n.a.	n.a.
Estonia	5.75	n.a.	n.a.
Finland	6	n.a.	n.a.
France	7 (vol.)	7	7
Germany	6.25	min. 4.4	min. 2.8
Greece	5.75	n.a.	n.a.
Hungary	5.75	4.4 (vol.)	4.4 (vol.)
Ireland	4 (vol.)	n.a.	n.a.
Italy	4	n.a.	n.a.
Latvia	5.75	5 (vol.)	5 (vol.)
Lithuania	5.75	n.a.	n.a.
Norway	5 (vol.)	n.a.	n.a.
Poland	6.2	n.a.	n.a.
Portugal	5	6.75 (vol.)	n.a.
Romania	5.75	5 (vol.)	5 (vol.)
Slovakia	5.75	min. 5.2 (vol.)	min. 3.1 (vol.)
Slovenia	5.5	n.a.	n.a.
Spain	7	min. 6	min. 3.9
Sweden	5.75	n.a.	n.a.
The Netherlands	4.25 (vol.)	3.5	3.5
UK	4 (vol.)	n.a.	n.a.

* Refers to the % of energy, unless otherwise specified (i.e. by volume)

Source: European Community for Sweet Sorghum and Ethanol

Appendix 3

Sweet sorghum characteristics (1/4)



- ▶ Sweet sorghum hybrids' vegetation period is 3 times shorter as compared to sugar cane's. Also, during the growing period, sweet sorghum needs 3 times less water than sugar cane, 2 times less than corn and up to 4 times less than other energetic crops.
- ▶ Compared to other agricultural crops, sweet sorghum has a much higher tolerance for parasites and other diseases.
- ▶ In irrigation conditions, it is possible to cultivate the sweet sorghum in the single-crop system.
- ▶ Sweet sorghum has an enormous potential for genetical improvement (there are currently over 4000 genetical forms - breeds, varieties, hybrids, etc).
- ▶ Sweet sorghum can be considered a crop of universal value, because it can be grown in all continents, in tropical, sub - tropical, temperate regions as well as in poor quality soils and semi-arid regions.
- ▶ Cultivation requirement and agronomic practice of sweet sorghum:
 - soil quality: acceptable heavy clay or high percent of sand;
 - soil acidity in a large diapason: pH 4,5 to 8,5;
 - soil salinity: better resistance than corn;
 - temperature: total accumulated daily temperature from 2600 grade Celsius up to 4600 grade Celsius during cycle with minimum germination temperature of 8 - 10 grade Celsius;
 - sweet sorghum is a plant for limited daylight hours;
 - water demand is 200 cubic meters/tonne with an average rainfall need of 500 to 600 mm.
- ▶ Sweet sorghum potential (medium term: 20 years):
 - the total world energy supply from high - energy sorghum and sugarcane could in a very long term (2050 and beyond) reach the amount equivalent to the 75% of the present world primary energy consumption;
 - photosynthetic efficiency (cca 4 g biomass/MJoules of solar radiation). Sweet sorghum is two times or more efficient in comparison with forest.

Appendix 3

Sweet sorghum characteristics (2/4)



Photo no.1. Sweet sorghum defoliated plants subjected to comparative tests. The Moldovan sweet sorghum hybrids, property of TARUS Intl. (no. 16-20) and sweet sorghum hybrids from research institutions of other countries (the description is in the basic text). Stalks 1 and 2 represent homologated Moldovan breeds.

Appendix 3

Sweet sorghum characteristics (3/4)



Photo no.2. Sweet sorghum plantation near Chisinau, Moldova

Appendix 3

Sweet sorghum characteristics (4/4)



Photo no.3. The latest harvest of the sweet sorghum hybrids (as of September 2011).

Results presented at the Munich Biomass Congress,
Oct 2006

Sorghum hybrid	Country	EtOH (l/ha)
Shennong 1	China	5600
Foralca	Italy	4200
Sofra	Italy	5000
Carmen	Romania	5100
Porumbeni 5	Moldova	6100
F.135 ST	Romania	6000



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